Trust-Tiered Knowledge Resolution and Adaptive Multi-Criteria Scoring for Localized Cold-Start Recommendations on Tasco Maps

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Local search and recommendation engines on digital map platforms are heavily constrained by spatial proximity, temporal availability, and database accuracy. In rapidly shifting urban ecosystems, keeping point-of-interest (POI) attributes fresh is a major operational challenge. According to the iPOS 2024 report, Vietnam has 323,010 food and beverage (F&B) outlets, but experiences a high churn rate with over 30,000 closures in a single half-year, making manual database curation (requiring 40 person-years per pass) unsustainable. This paper introduces a trust-tiered knowledge resolution and adaptive recommendation framework built for the Vietnamese F&B landscape on the Tasco Maps platform. The system integrates merchant point-of-sale data (MISA CukCuk) and vehicle dwell telemetry from the VETC mobility network (covering 2.8 million vehicles) into a five-tier registry. Conflicts are resolved using a consensus-based resolver ($N \geq 3$). For online serving, a sequential two-stage pipeline is implemented: (1) a Hard-Constraint Gate that enforces Vietnamese dietary preferences, spatial routing bounds, and opening hours to handle honesty traps, followed by (2) an Adaptive Multi-Criteria Soft-Scoring stage that switches between repeat and exploration travel modes and renormalizes weights for missing signals.

CCS Concepts: • **Information systems** → **Recommender systems**.

Additional Key Words and Phrases: Cold-Start Recommendation, Tasco Maps, VETC Telemetry, iPOS Vietnam F&B, Trust Resolver, Detour Decay, Weight Renormalization

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1 Introduction

Digital map platforms are essential for local discovery, navigation, and location-based commerce. While global platforms like Google Maps excel at global scale and breadth, they frequently suffer from lack of depth, unverified metadata, and black-box ranking in localized markets. This limitation is particularly prominent in the Vietnamese food and beverage (F&B) industry. According to the iPOS Vietnam 2024 F&B Market Report [1], the country has approximately 323,010 active outlets. However, the industry suffers from a high churn rate: over 30,000 outlets closed in the first half of 2024 alone, indicating a semi-annual churn rate of approximately 10%.

To keep a map platform's POI database accurate under such rapid decay, manual database curation is operationally unfeasible. At an estimated 15 minutes of human verification per POI, a single pass across 323,010 outlets requires 80,750 hours (equivalent to 40 person-years of manual labor). While large language models (LLMs) can scale extraction, naive LLM extraction frequently fabricates (hallucinates) missing attributes, causing brand damage.

This paper presents the design and formulation of a trust-tiered knowledge resolution and adaptive recommendation engine implemented for the next-generation digital map, **Tasco**

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Maps. The platform leverages two major local data alliances: (1) **MISA CukCuk** [2], a leading merchant POS provider in Vietnam with over 50,000 restaurants, and (2) **VETC** [3], a nationwide electronic toll collection network covering 2.8 million active vehicles. By combining merchant point-of-sale menus and real-time vehicle telemetry, the framework builds a self-updating, high-fidelity knowledge graph.

Our recommendation engine operates as a sequential pipeline: **Hard-Constraint Gating followed by Soft-Scoring**. The Hard-Constraint Gate prevents semantic-matching engines from silently relaxing constraints, addressing "honesty traps" (e.g., verifying if a requested dietary option actually exists). The Soft-Scoring stage ranks surviving candidates using a multi-criteria linear formulation incorporating geographic detour decay, POI quality, real-time occupancy, and localness priors. Additionally, the weights adapt dynamically based on travel frequency, separating repeat commuters from exploratory travelers, following the food delivery paradigm analyzed by Wang et al. [11].

2 Related Work

We build on cold-start POI recommendations, spatial detour modeling, and situation-aware systems.

2.1 Cold-Start Recommendation and LLM Priors

Collaborative filtering fails when items or users have zero historical interactions. Ju et al. [5] introduced the Language-Model Prior (LM-Prior) framework, demonstrating that zero-shot LLM representations of item metadata can establish a strong baseline prior. For specialized discovery, KALM4Rec [4] leverages retrieval-augmented LLM architectures to extract and verify query keywords. In Vietnamese contexts, morphological segmenters like HuTieuBERT by Dinh et al. [6] are critical for parsing localized dish names (e.g., "bún chả") and preventing entity-matching failures. Furthermore, processing noisy social media data (e.g., reviews, slang, and dialect expressions extracted from platforms like TikTok or Threads) requires robust domain-specific language models. For this purpose, we leverage ViSoBERT proposed by Nguyen et al. [14], a pre-trained Transformer model optimized for Vietnamese social media text, enabling accurate aspect-based sentiment extraction and linguistic normalization in our offline enrichment pipeline.

2.2 Geographical POI Recommendation

POI recommendation is fundamentally shaped by geography. GeoMF by Lian et al. [7] models spatial regions within matrix factorization by applying spatial decay. In highway or corridor environments, routing recommenders must account for detour paths rather than straight-line distance. Dynamic spatial routing structures (e.g., GETNext by Yang et al. [8]) model trajectory transitions, but their parameter size causes overfitting on small localized datasets, highlighting the need for parametric decay functions that scale with detour times.

2.3 Situation-Aware Repeat vs. Explore Recommendations

User preferences fluctuate between familiar routines and novel exploration. Wang et al. [11] studied food consumption patterns and proposed dual-model architectures (RepRec and ExpRec) to separate repeat orders from exploratory choices, showing that unified recommendation pipelines underperform when situation-aware contexts are ignored.

3 System Architecture and Preference Channels

The system separates offline background knowledge resolution from online real-time serving, ensuring low latency and resilience during network drops. The architecture features a five-tier registry that ingests and resolves data through a multi-trigger loop.

3.1 Multi-Trigger Lifecycle Loop

Rather than treating the database as static, the system runs a multi-trigger lifecycle loop governed by a set of triggers $\mathcal{T} = \{T_1, T_2, T_3, T_4, T_5, T_6\}$:

- **Bootstrap (T_1):** Initializes the database by ingesting baseline files and OpenStreetMap (OSM) nodes.
- **Gap-Detect (T_2):** Triggered when a POI's quality score $Q < 0.75$ or key attributes (e.g., operating hours) are absent. This triggers a localized agent run to enrich the profile.
- **TTL Expiry (T_3):** Triggered when the age of a resolved field exceeds its tier-specific Time-To-Live (τ), prompting a target refresh of the expired source: $\tau_1 = 180$d (Tier 1), $\tau_2 = 90$d (Tier 2), $\tau_3 = 30$d (Tier 3), $\tau_{4a} = 14$d (Tier 4a).
- **Business Event (T_4):** Triggered when a merchant uploads a menu image, initiating zero-shot OCR and structured menu extraction.
- **Driver Telemetry (T_5):** Triggered when GPS telemetry detects a vehicle passing and dwelling at a POI for ≥ 25 minutes, prompting a 1-tap validation.
- **User Query (T_6):** Triggers the online search and recommendation pathway.

3.2 Preference Input Channels

To construct personalization models with minimal user effort, the system aggregates user preferences through four complementary channels:

(1) **Trip Context (CTX):** Fully passive channel extracting context from the active route navigation state, including the timestamp (t_{now}), arrival ETA (t_{arrival}), and vehicle class (e.g., cars require parking).
(2) **Spatial-Temporal Autocomplete:** Captures query intents based on character prefixes combined with spatial-temporal priors. For example, the query prefix "ca" suggests coffee shops ("cà phê") at 7:00 AM in urban zones, but shifts to fish soup ("cơm cá") at 12:00 PM on a highway corridor. This addresses the 17.9% intent variance observed by Li et al. [12] (Baidu MST-PAC).
(3) **Onboarding Chips:** A one-time 10-second setup capturing critical hard constraints (allergens, dietary restrictions, or family travel) where incorrect predictions pose safety or high-dissatisfaction risks.
(4) **Implicit Learned Behavior:** Collects telemetry-based signals such as vehicle dwell times (indicating repeat customers), navigation requests, and post-trip 1-tap feedback.

4 Trust-Tiered Knowledge Resolution

Data is ingested from multiple sources, each mapped to a specific trust tier $t \in \{1, 2, 3, 4a, 4b\}$, as shown in Table 1. Each tier carries a static confidence weight $C_t \in [0, 1]$ and a time-to-live τ_t (in days).

Let $\mathcal{V}_f = \{(v, t, \Delta t)_j\}$ be the set of candidate values extracted for a field f from active sources, where v is the value, t is the source's trust tier, and Δt is the data age. The Resolver determines the published value v_f^* and its confidence $\text{conf}(f)$ through the following consensus rules:

Table 1. Trust Tier Hierarchy

Tier	Source Type	Confidence (C_t)	TTL (τ_t)
Tier 1	Merchant Verified (MISA CukCuk POS / OCR)	0.95	180 days
Tier 2	Driver Telemetry (VETC vehicle dwells)	0.85	90 days
Tier 3	Licensed API (OSM/Tasco CSV)	0.80	30 days
Tier 4a	Agentic Social Scraping (Foody/Riviu)	0.60	14 days
Tier 4b	Social Buzz Signals (TikTok/Threads)	N/A	Decaying

(1) **TTL Filtering:** Remove all candidates where the elapsed time $\Delta t > \tau_t$.
(2) **Highest Tier Selection:** Select candidates corresponding to the highest surviving tier $t_{\max}$. If multiple distinct values exist within the same tier, the most recently updated value (lowest Δt) is selected.
(3) **Consensus Constraint:** For Tiers 2 (driver confirmed) and 4a (agentic facts), a candidate value is only valid if supported by a consensus of $N \geq 3$ independent sources. A single agentic source cannot override fresh licensed data.
(4) **Absence Gating:** If no candidates survive, the field is marked as *absent*. No default values are fabricated. This prevents recommendation hallucinations under strict search queries (such as queries for non-existent entities like "Crystal BBQ").
(5) **Full History Retention:** Retains the entire candidate list to provide source provenance for the RAG assistant (/assistant).

The resolver preserves the winning tier's confidence as $\mathrm{conf}(f) = C_{t_{\max}}$, which is propagated to the ranking stage. Tier 4b sources bypass this resolver and feed directly into the soft scoring stage as transient popularity weights.

5 Mathematical Formulation and Online Serving

Let $U = (q, l, s, d, p_c, t, \mathcal{H})$ represent the user context, where q is the query string, $l = (\phi_u, \lambda_u)$ is the location, $s \in \mathcal{S}$ is the user segment, $d \in \mathcal{D}$ denotes dietary requirements, p_c is the maximum price threshold, t is the timestamp, and $\mathcal{H}$ is the user's travel history on the current route. Let R denote a candidate restaurant POI.

5.1 Stage 1: Hard-Constraint Gating

To prevent semantic matching from violating physical constraints, candidate POIs are filtered using a binary indicator $H(R, U) \in \{0, 1\}$:

$$H(R, U) = \prod_{k \in \mathcal{K}_{\text{active}}} h_k(R, U) \tag{1}$$

where $\mathcal{K}_{\text{active}}$ represents the active constraint set, evaluating:

- **Dietary Match ($h_{\mathbf{diet}}$):** Checks if dietary tags match the resolved properties of R or its menu items:

$$h_{\text{diet}}(R, U) = \begin{cases} 1 & \text{if } d = \emptyset \text{ or } \exists m \in \mathcal{M} \text{ s.t. } \mathrm{status}(m, d) \geq 0.60 \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

- **Geographic Max Boundary ($h_{\mathbf{geo}}$):** Validates if the physical distance is within the query radius limit.

- **Operating Hours (h_{time}):** Evaluates opening hours at the estimated arrival time $t_{\text{arrival}} = t_{\text{now}} + \text{ETA}$, rather than the query time t_{now}.
- **Price Ceiling (h_{price}):** Rejects R if the average price $P_a > p_c$ or if all matched dishes exceed p_c.
- **Parking Requirement (h_{parking}):** Escalated to a hard constraint if vehicle class is set to 'car'.

If the gated set $\mathcal{R}_{\text{gate}} = \{R \mid H(R, U) = 1\} = \emptyset$, the system returns an explicit empty state rather than relaxing constraints silently, addressing honesty traps (such as requests for Halal food in districts with zero certified options).

5.2 Stage 2: Multi-Criteria Soft-Scoring

For each candidate $R \in \mathcal{R}_{\text{gate}}$, we compute the unified score:

$$\tilde{S}(R, U) = \sum_{i \in \mathcal{I}} w_i S_i(R, U) \tag{3}$$

where $\sum_{i \in \mathcal{I}} w_i = 1$, and each $S_i(R, U) \in [0, 1]$. The components are:

(1) **Semantic Relevance (S_{semantic}, $w_1 = 0.30$):** Cosine similarity of embeddings plus Jaccard token similarity over normalized unigrams.
(2) **Preference Match (S_{match}, $w_2 = 0.20$):** Matches query preferences against POI tags, scaled by the resolved field confidence:

$$S_{\text{match}}(R, U) = \text{MatchRatio}(q, \mathcal{A}) \cdot \text{conf}(\mathcal{A}) \tag{4}$$

where $\text{conf}(\mathcal{A})$ is the confidence of the attributes generated by the Resolver.
(3) **Detour Decay (S_{route}, $w_3 = 0.15$):** Modeled using exponential decay over detour time:

$$S_{\text{route}} = \exp\left(-\frac{\Delta t_{\text{detour}}}{\lambda}\right) \tag{5}$$

where Δt_{detour} is the detour time in minutes and $\lambda = 5$ minutes.
(4) **Behavioral History (S_{behavior}, $w_4 = 0.15$):** Extracted from vehicle telemetry $\mathcal{H}$ representing historical visit frequency on the current corridor.
(5) **Information Quality (S_{qual}, $w_5 = 0.10$):** Static completeness metric $Q/100$.
(6) **Crowd Occupancy (S_{crowd}, $w_6 = 0.05$):** Real-time congestion index:

$$S_{\text{crowd}} = 1 - \frac{\text{occupancy}(t_{\text{arrival}})}{100} \tag{6}$$

(7) **Social Buzz (S_{buzz}, $w_7 = 0.05$):** Decaying trending score from Tier 4b sources. Textual features from social posts are parsed using ViSoBERT [14] to extract aspect-based sentiments before computing the transient buzz index.

5.3 Adaptive Renormalization for Missing Signals

Online requests often lack GPS context (denied permissions) or behavioral logs. To prevent score distortion, we exclude missing features from the active set $\mathcal{I}_{\text{active}} \subseteq \mathcal{I}$ and renormalize the weights:

$$S(R, U) = \sum_{i \in \mathcal{I}_{\text{active}}} w_i^* S_i(R, U) \quad \text{where} \quad w_i^* = \frac{w_i}{\sum_{j \in \mathcal{I}_{\text{active}}} w_j} \tag{7}$$

5.4 Context-Aware Mode Switching

Following the repeat-explore behavioral dichotomy in local food delivery studied by Wang et al. [11], we implement dynamic mode-switching based on route familiarity. A route is classified as *familiar* if user history contains ≥ 3 trips within the past 90 days. The weights are dynamically switched as follows:

- **Repeat Mode (Familiar Routes):** Prioritizes user habit and crowd status, decreasing the semantic weight and ignoring temporary social buzz:
$$w = [0.25, 0.20, 0.15, 0.25, 0.10, 0.05, 0.00]$$
representing $[w_{\text{sem}}, w_{\text{mat}}, w_{\text{rout}}, w_{\text{beh}}, w_{\text{qual}}, w_{\text{cro}}, w_{\text{buz}}]$.
- **Explore Mode (New Routes):** Prioritizes semantic discovery and trending social buzz:
$$w = [0.35, 0.20, 0.15, 0.05, 0.10, 0.05, 0.10]$$

5.5 Vietnam-Specific Adjustments: Localness and Anti-Luxury Bias

Under the Tasco Map platform guidelines, there is a preference for small, local, and authentic eateries ("bình dân") over high-end international luxury chains. To incorporate this, the final score incorporates a localness adjustment:

$$S_{\text{final}}(R, U) = S(R, U) + \alpha \cdot \text{Localness}(R) - \beta \cdot \text{Luxury}(R) \tag{8}$$

where $\text{Localness}(R) = 1$ is triggered if the price tier is budget and POI quality is high, and $\text{Luxury}(R) = 1$ is triggered if the price level is premium. The hyperparameters α and β control the strength of this localization prior.

5.6 LLM Rerank and Explainer

Following the classical-plus-neural recommendation architectures described by DoorDash [13], the top 20 candidates generated by Stage 2 are passed to an LLM Reranking model. The classical scoring model manages scaling and candidate pruning, while the LLM incorporates contextual semantics and generates natural language explanations (e.g., "4 minutes detour from route, has car parking [owner verified], positive reviews on TikTok") in Vietnamese.

6 Discussion and Future Work

The trust-tiered consensus resolver provides a robust mechanism to manage the trade-off between coverage and accuracy. While scraping social platforms increases coverage, it introduces noise. The consensus constraint ($N \geq 3$ for Tier 4a) acts as an automated filter, preventing unverified claims from contaminating the knowledge base. Furthermore, propagating the resolver's confidence score ($\text{conf}(f)$) into S_{match} ensures that highly verified facts naturally rank higher than agentic guesses.

Currently, the weights for Repeat and Explore modes are statically defined. A logical next step is to train a meta-learning ensemble layer to interpolate weights based on continuous context features. Additionally, while the linear model is highly suitable for cold-start environments due to its interpretability, scaling to millions of POIs will necessitate a deep architecture. In such scenarios, the Hard-Constraint Gate will continue to serve as a pre-filtering layer, maintaining system honesty.

7 Conclusion

We presented a two-stage localized recommendation framework that integrates trust-tiered knowledge resolution and multi-criteria soft-scoring. By layering a consensus resolver over merchant, telemetry, and agentic sources, we establish a verifiable foundation for query evaluation. The sequential integration of hard-constraint gating, detour-based geographic decay, and context-aware mode switching balances spatial routing requirements and user preferences. The adaptive weight renormalization rule ensures robust execution under sparse signals, presenting a production-ready approach for next-generation map recommendation systems in Vietnam.

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